

Spotify Track Popularity Prediction

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Github repo: <https://github.com/powerj05/spotify-analysis>

1. Introduction

Music streaming platforms like Spotify host millions of tracks, and every day thousands of new songs compete for listener attention. Understanding what makes a song popular is a commercially valuable question, and is highly relevant to algorithmically driven platforms such as Spotify.

It is common for algorithms to attempt to predict the popularity of a song using various features describing the song itself. These can be objective measures such as loudness, tempo or key, or subjective qualities as determined either by human judgement or machine-learning algorithms. Spotify in particular provides a number of interesting measures of the various qualities of the songs on its platform through its Web API, including variables such as “energy”, “acousticness” and “danceability”, as well a score of popularity from 0 to 100. However, while these features may be useful in describing the sound of the song itself, these are not the only factors that drive a song’s popularity. In practice, a song's commercial success is often influenced heavily by other factors such as marketing, and the established popularity of its

artist. Despite their relevance, these factors are much less often included in models of the popularity of songs.

In order to investigate the significance of these factors, this project uses various modelling techniques to predict the popularity of songs on Spotify both with and without the use of measures of the popularity of the artist, such as follower counts and monthly listeners. A key focus of this analysis is quantifying the "Fame Premium", which is the extra popularity a song receives simply because of the artist's market power. By breaking down these variables, this report aims to show the real factors behind a song's commercial success.

1. Methodology

2.1 Data Description

The project initially aimed to use the most recent data possible by pulling data from the [Spotify Web API](#) to create a dataset. However, the API has become much more restricted recently, and it was not possible to obtain data on artists alongside their songs. Data were instead obtained from a [Kaggle dataset](#) containing information pulled from the Spotify Web API about both songs and artists featured on Spotify Editorial playlists in April and May 2024. In total 15,000 unique tracks and 10,000 unique artists were included in the dataset across two separate files.

The track data included the following variables, in addition to the name and unique Spotify ID of each track and other variables not considered relevant for the research:

- popularity - Spotify's metric for the popularity of the song; this was the target variable for the models used (see below)
- markets - the number of markets in which the track is available
- artists - the unique Spotify IDs of each artist who created the track
- release_date - the year or date on which the song was released

The following descriptions of the audio features of each song are summarised from the Spotify Web API documentation:

- acoustictness - a confidence measure from 0.0 to 1.0 of whether the track is acoustic. 1.0 represents high confidence the track is acoustic.
- danceability - describes how suitable a track is for dancing based on a combination of musical elements including tempo, rhythm stability, beat strength, and overall regularity. A value of 0.0 is least danceable and 1.0 is most danceable.
- energy - a measure from 0.0 to 1.0 and represents a perceptual measure of intensity and activity. Typically, energetic tracks feel fast, loud, and noisy. Perceptual features contributing to this attribute include dynamic range, perceived loudness, timbre, onset rate, and general entropy.
- instrumentalness - predicts whether a track contains no vocals. The closer the instrumentalness value is to 1.0, the greater likelihood the track contains no vocal content. Values above 0.5 are intended to represent instrumental tracks, but confidence is higher as the value approaches 1.0.
- key - the key the track is in. Integers map to pitches using standard [Pitch Class notation](#). E.g. 0 = C, 1 = C#/Db, 2 = D, and so on. If no key was detected, the value is -1.
- liveness - detects the presence of an audience in the recording. Higher liveness values represent an increased probability that the track was performed live. A value above 0.8 provides strong likelihood that the track is live.

- loudness - the overall loudness of a track in decibels (dB). Loudness values are averaged across the entire track and are useful for comparing relative loudness of tracks. Values typically range between -60 and 0 db.
- mode - indicates the modality (major or minor) of a track, the type of scale from which its melodic content is derived. Major is represented by 1 and minor is 0.
- speechiness - detects the presence of spoken words in a track. The more exclusively speech-like the recording (e.g. talk show, audio book, poetry), the closer to 1.0 the attribute value. Values above 0.66 describe tracks that are probably made entirely of spoken words. Values between 0.33 and 0.66 describe tracks that may contain both music and speech, either in sections or layered, including such cases as rap music. Values below 0.33 most likely represent music and other non-speech-like tracks.
- tempo - the overall estimated tempo of a track in beats per minute (BPM). In musical terminology, tempo is the speed or pace of a given piece and derives directly from the average beat duration.
- time_signature - an estimated time signature; ranges from 3 to 7 indicating time signatures of "3/4", to "7/4".
- valence - a measure from 0.0 to 1.0 describing the musical positiveness conveyed by a track. Tracks with high valence sound more positive (e.g. happy, cheerful, euphoric), while tracks with low valence sound more negative (e.g. sad, depressed, angry).

Many of the above variables can be objectively measured (key, tempo, loudness, etc.); however, most of them are more subjective in nature and cannot be exactly quantified, such as energy, valence, and danceability, or are estimates of confidence of an objective quality, such as acousticalness or liveness. It is important to note that Spotify does not share how these variables are calculated; it seems likely that they are largely predicted by a machine-learning algorithm. Of the target variable, Spotify's measure of popularity, it is known only that it is "based, in the most part, on the total number of plays the track has had and how recent those plays are."

This element of recency has important implications for the results of the analysis presented here: because the data were all gathered in 2024, the popularity of songs may be a result of specific trends at the time, and the extensibility to other time periods of the models explored here may therefore be limited. However, the data provided is sufficient to investigate the premise of the research, as it still provides ample opportunity to explore the relationship between the various features and popularity.

The artist dataset included the following data on each artist, in addition to name, ID, and data relating to their appearances on Spotify playlists:

- followers - number of followers on Spotify
- genres - the different genres attributed to an artist; multiple genres were stored as comma-separated strings
- first_release - year of artist's first release
- last_release - year of artist's most recent release as of May 2024
- num_releases - total number of releases by artist
- num_tracks - number of tracks on artist's most recent release

2.2 Preprocessing

To ensure the reliability of our models, it is important to clean the raw dataset. First, we addressed any missing values and removed duplicate records to maintain data integrity. During analysis, we identified market metrics, such as an artist's monthly listeners and follower count. They show highly skewed distributions with significant outliers. To mitigate this without losing valuable indicators, we applied logarithmic transformations to these variables (creating features like `log_monthly_listeners` and `log_followers`).

The exploratory data analysis provides an overview of the distribution of key variables and their relationship with track popularity. The target variable, popularity, is approximately normally distributed, with most values concentrated between 50 and 70, alongside a noticeable cluster at zero, indicating tracks with minimal exposure.

Audio features generally present smooth distributions but show weak relationships with popularity. While danceability shows a slight positive trend, variables such as energy and valence display no clear association with popularity. Song duration is concentrated between 2 and 4 minutes and does not appear to significantly influence performance. In contrast, market-related variables demonstrate much stronger relationships. Artist popularity, follower counts, and monthly listeners all show clear positive correlations with track popularity, with monthly listeners appearing to have the strongest association. The correlation matrix further confirms that market variables are more strongly correlated with popularity than audio features.

Overall, the EDA suggests that while audio features describe the intrinsic qualities of a song, they have limited explanatory power. Instead, market factors such as audience reach and artist visibility play a dominant role, providing initial support for the concept of a “fame premium.”

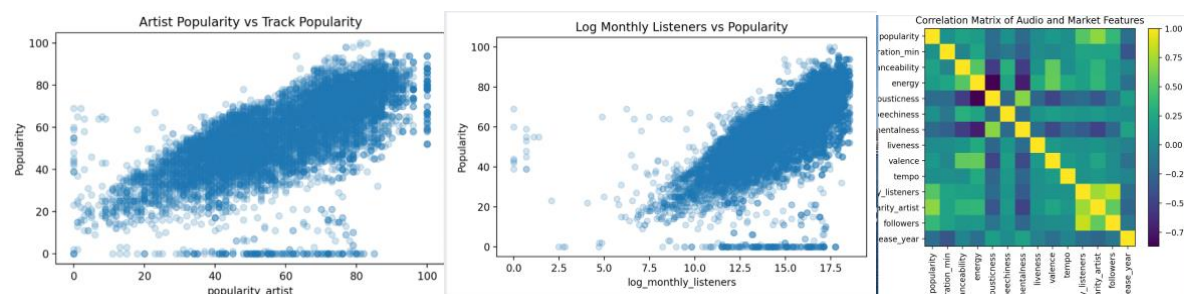


Figure 1. Artist Population vs. Track Popularity, Log Monthly Listeners vs. Popularity, Correlation Heatmap

2.3 Modeling Strategy

To isolate and quantify the impact of artist fame on a song's success, we designed a comparative modeling strategy using two parallel models. Model 1 (audio only) was trained exclusively on the 13 audio features listed above. This model works as a baseline, capturing only the basic sonic properties of a track. Model 2 (Audio + Market) extends Model 1 by adding artist-level features (e.g., artist popularity, log-transformed follower counts) to determine the added predictive power of an artist.

To evaluate model performance on unseen data, the cleaned dataset was partitioned into an 80% training set and a 20% testing set. And the model performance was mainly evaluated using R^2 and Mean Absolute Error (MAE).

2.3 Linear Regression

Before deploying advanced ensemble methods, we first implemented standard Linear Regression to set up a predictive baseline. This classical statistical approach assumes a linear relationship between the independent variables (audio and market features) and the dependent variable (popularity). By evaluating this baseline, we aimed to test whether a song's success scales linearly with its attributes, or if the underlying market dynamics require more complex, non-linear modeling. The model performance was evaluated using MSE, MAE, and R^2 .

2.4 Random Forest

The random forest method makes use of bootstrapping to train a large number of decision trees on different samples of the data, with additional randomisation employed in the formulation of the trees to ensure sufficient diversity throughout the “forest”. Predictions are then made by running a sample through all of the trees and taking the average result. This project made use of the random forest model provided by the Scikit-learn (sklearn) package for Python.

In order to utilise the categorical variables in the dataset, some additional preprocessing steps were undertaken to prepare for the random forest model. First, the `time_signature` variable was deconstructed into binary dummy variables. Second, the `genre` variable had to be “target encoded” - the string label for the genre of each song was replaced with the numerical value of the average popularity of all songs of that genre. Where no genre was present, songs were assigned an “unknown” value for genre which was target-encoded separately, and where multiple genres were present, only the first was taken for ease of calculation. As with the other methods employed, an 80/20 training/test split was used to develop the model.

The model selection involved determining the optimal combination of five parameters. This was done using a randomised search using five-fold cross-validation: thirty random combinations of pre-defined values for the parameters were tested using five-fold cross validation, and the combination with the lowest Root Mean Squared Error was selected for use. After running a model that included all relevant artist features in addition to track data, this method was repeated with the artist features omitted to provide a comparison. In order to assess how the importance of features changed between models, the `permutation_importance` method from `sklearn.importance` was used to calculate how much the RMSE worsens when the values of each feature are randomly shuffled across the sample.

2.5 Gradient Boosting

While linear regression provides a baseline for understanding general relationships, we utilized Gradient Boosting to capture more complex, non-linear interactions and to further analyse prediction behaviour.

In addition to evaluating raw predictive performance (besides R^2 and MAE), we categorized the regression predictions into three popularity tiers: Low (< 40), Mid (40–70), and High (> 70), assessing them using accuracy, precision, recall, and weighted F1-scores. Also we defined a “Fame Premium” metric to isolate the impact of market factors, it calculated as actual popularity minus the audio only predicted popularity. Finally, SHAP analysis was used on Model 2. Features are ranked vertically by overall importance, while each dot represents a single song. Its position on the x-axis shows whether it increased or decreased the predicted popularity, and its color indicates the actual value of that feature (red for high, blue for low). Its values quantify the marginal contribution of each feature to each individual prediction, visualize and quantify the hierarchy of feature importance.

2. Results & Analysis

3.1 Linear Regression

The audio-only model shows very weak predictive power ($R^2 = 0.0406$), with high error (MAE = 14.23), indicating that audio features alone cannot explain popularity. In contrast, the audio + market model performs significantly better ($R^2 = 0.4445$), with reduced error (MAE = 9.91), representing an approximate 30% improvement. These results suggest that music popularity

is driven more by market factors than intrinsic audio characteristics, supporting the idea of a “fame premium.”

However, linear regression assumes simple linear relationships, it may fail to capture complex patterns in music popularity.

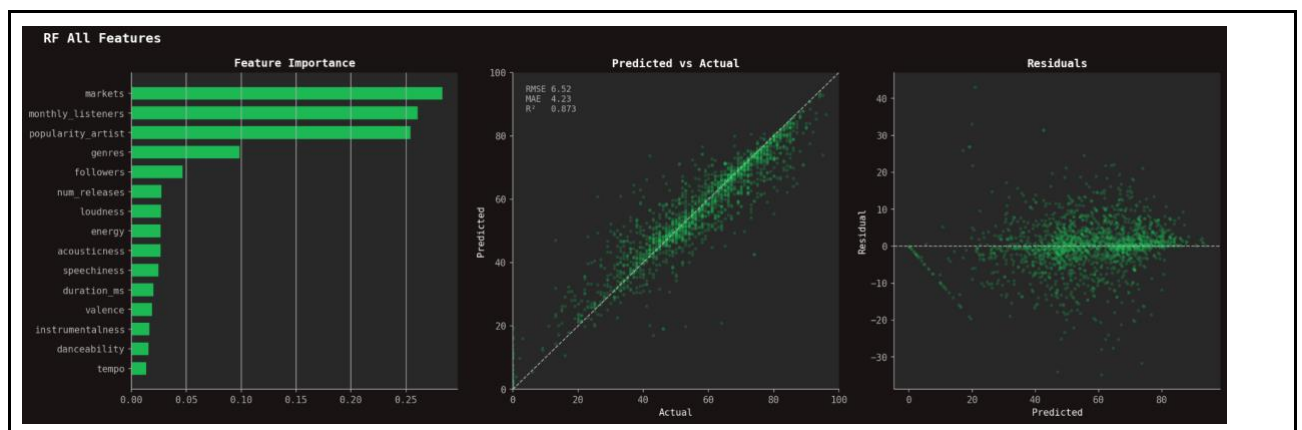
3.3 Random Forest

The optimal random forest model produced with the randomised search had the following parameters:

Parameter	Explanation	Value
n_estimators	Number of decision trees to produce	500
max_depth	Maximum depth of decision tree/maximum number of splits to make in the data	30
min_samples_split	Minimum number of samples required at a node to split	2
min_samples_leaf	Minimum number of samples at each leaf node	1
max_features	Number of randomly selected features to consider for splitting at each node (proportion of total # features)	0.33

These features allow for very deep, complicated decision trees. This would normally cause the model to severely overfit to the training data, indicated by a significantly larger error on the test set; however, the 6.518 RMSE on the test set was actually lower than the average RMSE of 6.838 from the 5-fold CV on the training set. The model also achieved a Mean Average Error (MAE) of 4.231 and R^2 of 0.873, significantly better performance than the linear regression or gradient boosting models.

As expected, the artist data was the most important in determining the popularity of songs - monthly listeners, follower count, and artist popularity all featured in the top five most important features alongside genre. Unusually, however, the most important feature in both models was the number of markets that the song was available in.



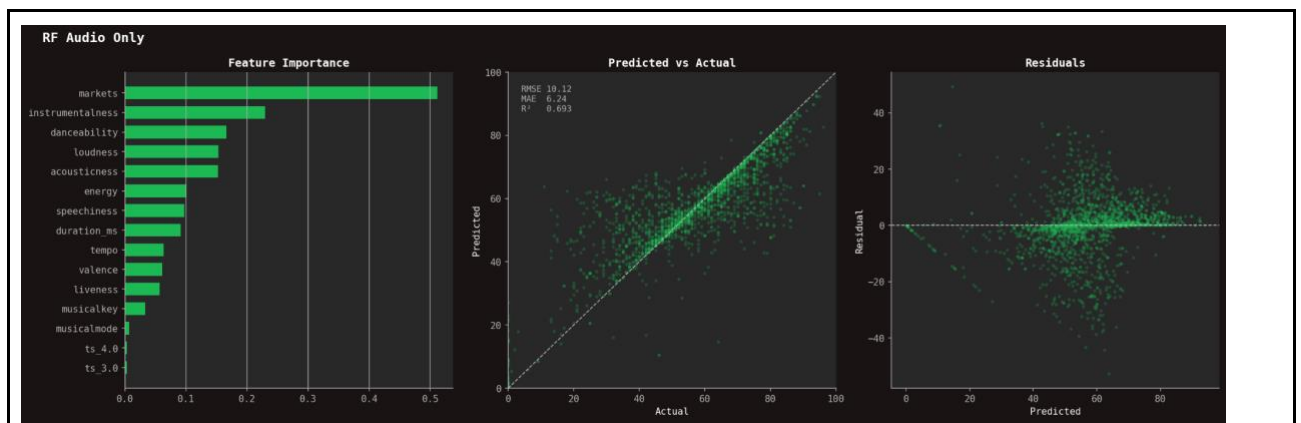


Figure 2. Diagnostics of random forest models

However, despite the importance of artist popularity data in the first model, removing these features did not worsen the model's performance by as much as expected, with the track-only model still producing relatively low MAE of 6.5. Nonetheless, the model's performance was still worse with this specification, as the predicted values deviated significantly from actual values further from the mean.

	Audio only	Audio + Artist data
RMSE	6.518	10.121
MAE	4.231	6.237
R ²	0.873	0.693

3.4 Gradient Boosting

The performance gap between the two models is also substantial in the gradient boosting model. The Model 1(audio only) explains only about 8% of the variance in popularity and shows an average prediction error of about 14. When market features are added in Model 2, performance improves dramatically, R² rises to 0.529 and MAE falls to 8.59 which is a 37% reduction in error.

	R ²	MAE
Model 1 (Audio Only)	0.084	13.57
Model 2 (Audio + Market)	0.53	8.59

+0.45

-36.7%

Figure 3. Regression Performance Comparison

The confusion matrices illustrate the difference between the two models. Model 1 can hardly distinguish every tiers, it has a recall of 1.00 for the Mid class and classifies almost every song as Mid-tier. Model 2 performs far better across all three classes, achieving an overall accuracy of 72% and a weighted F1 of 0.71. It correctly identifies 88% of Mid-tier songs and has weaker performance on High- and Low- tier songs (44%), meaning many genuinely obscure songs are expected to be more popular. This asymmetry reflects a tendency in the model to overestimate popularity for songs by popular artists, a pattern we tested further in the error analysis.

The classification reflects that there's still a lot of noise and randomness involved in the entire prediction. Relying only on audio features, the model cannot capture market signals (such as

the artist’s own fame, fan base, and external reach). Therefore, it can only make predictions based on the acoustic characteristics of the music itself.

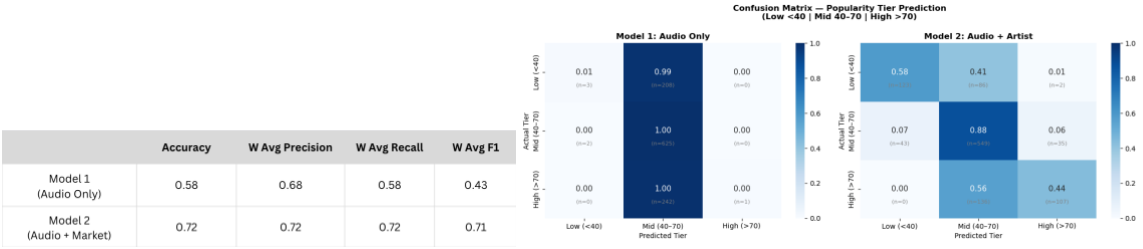


Figure 4. Classification Performance Metrics and Confusion Matrices for Popularity Tier

The model does not make errors randomly, they follow systematic patterns. Looking at error by actual popularity tier, the model overestimates popularity for low-tier tracks (mean bias of -11.5, indicating the model predicts too high) and underestimates for high-tier tracks (mean bias of +8.9, indicating the model predicts too low). This shows it is regression-to-the-mean, which is common in machine learning models trained on unbalanced or noisy targets. Thus the model is less useful precisely at the extremes where decisions are most consequential.

Looking at error by artist popularity tier, the model is most accurate for songs by low-popularity artists (MAE 5.54) and less accurate for songs by mid- and high-popularity artists (MAE 9.71 and 9.32). This counterintuitive finding is driven by the extremely low variance in the success of unknown artists, whose tracks predictably remain at the lower end of the scale. In contrast, popular artists produce more variable outcomes, and the model cannot reliably distinguish between these cases based on artist metrics alone.

We noticed that adding artist features improved overall accuracy, but in some cases it also caused the model to rely too heavily on the artist's fame. For example, 'Make You Feel My Love', where the model predicts a popularity score of 69, the artist's popularity 81, but the actual score is 0.

Actual Popularity Tier	MAE	Bias	Std Error	Artist Popularity Tier	MAE	Bias	Std Error
Low (<40)	13.245	-11.526	17.654	Low (0-40)	5.538	0.202	7.996
Mid (40-70)	6.684	0.664	8.55	Mid (41-70)	9.71	-0.25	14.213
High (>70)	9.466	8.864	7.234	High (71-100)	9.32	0.774	12.9

Figure 5. Error Analysis by Actual and Artist Popularity Tiers

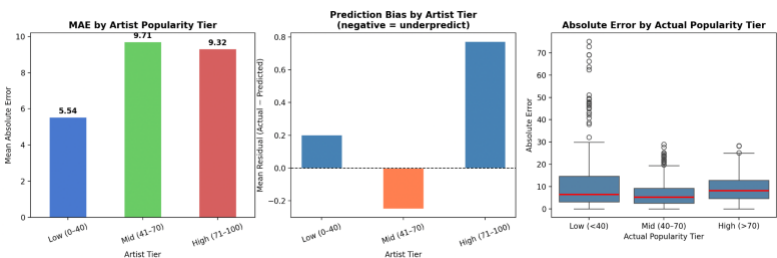


Figure 6. Prediction Error and Bias Distribution (Model 2)

The fame premium distribution is roughly centred at 0 but slightly right-skewed, with a median of 1.2 and 52.9% of songs having a positive value. This means that just over half of songs have a popularity boost from market factors beyond what their audio would predict. The scatter plot of fame premium against artist popularity score tells us an asymmetric pattern, artists with mid-range popularity scores (40–70) show the widest spread of fame premiums in both directions. This suggests that this is where market outcomes are most unpredictable. Very popular artists tend to have songs that cluster closer to zero premium, their popularity is expected by the model and therefore already priced in. Conversely, some songs by unpopular artists achieve fame premiums well above +20, suggesting genuine undervalued tracks with strong audio potential that have not yet found their audience. Indicating that market power

often causes a song's popularity to exceed what can be explained by its audio characteristics alone.

Spotify Popularity — Gradient Boosting Full Analysis



Figure 7. Fame Premium Distribution and Artist Popularity

Figure 8. SHAP Summary Plot for Model 2

SHAP analysis of Model 2 presents a clear hierarchy among features. The most important predictor is `log_monthly_listeners` and followed by `popularity_artist`. These are all market factors, explain the overwhelming majority of the model's predictive power.

3. Model Comparison

Overall, all three models proved the hypothesis that the popularity of an artist is a significant determinant of the popularity of one of their songs, much more so than any qualities of the song itself. All models made large gains in R^2 after artist popularity data were included, and MAE also reduced accordingly across all models.

Model	R^2	MAE
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Linear Regression	0.04/0.44	14.23/9.91
Random Forest	0.693/0.873	6.237/4.231
Gradient Boosting	0.084/0.53	13.57/8.59

However, of the three, the random forest model stands out as by far the most accurate at predicting popularity, even before an artist's following is accounted for. This is likely due to the large number of deep, complicated decision trees used in this model which allow it to complex, non-linear relationships between features and maximise information gain.

4. Conclusion & Discussion

This project aimed to investigate the determinants of a song's popularity using data on the artist's popularity in addition to its audio qualities with a view to determining how important the former is to a song's success. All models used to explore this question proved the initial hypothesis that the size of an artist's following is a significant predictor of the success of their songs, and a very high predictive power was achieved using random forest models. However, this project has a number of limitations that should be addressed by further research.

Firstly, the use of data from a specific point in time (April-May 2024) limits the external validity of the models presented here. Despite the high accuracy of the models, it is important to recall that Spotify's internal metric for popularity is weighted by the recency of a song's plays in addition to the number of listeners. These models, therefore, may be able to accurately predict how popular a song would have been in April to May of 2024, but may fail to capture changing trends if applied to data from a different time period; they could therefore be made more relevant by training on the most recent data available.

Another key limitation of the project is its use of cross-sectional data, as it fails to observe changes in both song popularity and artist following over time. This makes it difficult to identify the direction of causality, which is an important distinction to make in this context: while a song by an already popular artist may be likely to do better as a result, it is also possible that a song by an unknown artist may cause that artist to significantly in popularity. An expansion of the dataset to incorporate a time element could therefore significantly improve the accuracy of its predictions.